

Quiz 4

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Packages

Our usual `tidyverse` package will be enough for this quiz.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Data—read this!

We will use a data set with information about 110 car models from the 2020 model year from Consumer Reports, a publication that offers buying advice for many products including cars. The authors of another statistics textbook collected these data from the Consumer Reports web site (<https://www.consumerreports.org/cars/>).

The original data set has many variables, but I have kept only two:

- **Weight**, which is the weight of the car in pounds, and
- **CityMPG**, which is the number of miles that the car can drive on city streets on one gallon of gas, according to Consumer Reports's tests.

In loading the data, I corrected a couple of mistakes in the data set. (The weight for the Buick Regal was mistyped as 2680 instead of 3680. The CityMPG value for the Audi A8 was clearly incorrect, but I could not find the correct value, and so I dropped the entire row from the data set.)

I also created a new variable `weight_1000`, which is the weight of the car measured in thousands of pounds. I created this new variable to make a slope that you will find more meaningful.

```
cars_2020 <- read_csv("http://www.lock5stat.com/datasets3e/Cars2020.csv") %>%
  select(Make, Model, Weight, CityMPG) %>%
  rows_update(tibble(Model = "Regal", Weight = 3680)) %>% filter(Model != "A8") %>%
  mutate(weight_1000 = Weight / 1000)
```

```
## Rows: 110 Columns: 21
## -- Column specification -----
## Delimiter: ","
## chr   (5): Make, Model, Type, Drive, Size
```

```
## dbl (16): LowPrice, HighPrice, CityMPG, HwyMPG, Seating, Acc030, Acc060, Qtr...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Matching, by = "Model"
```

```
head(cars_2020, 100)
```

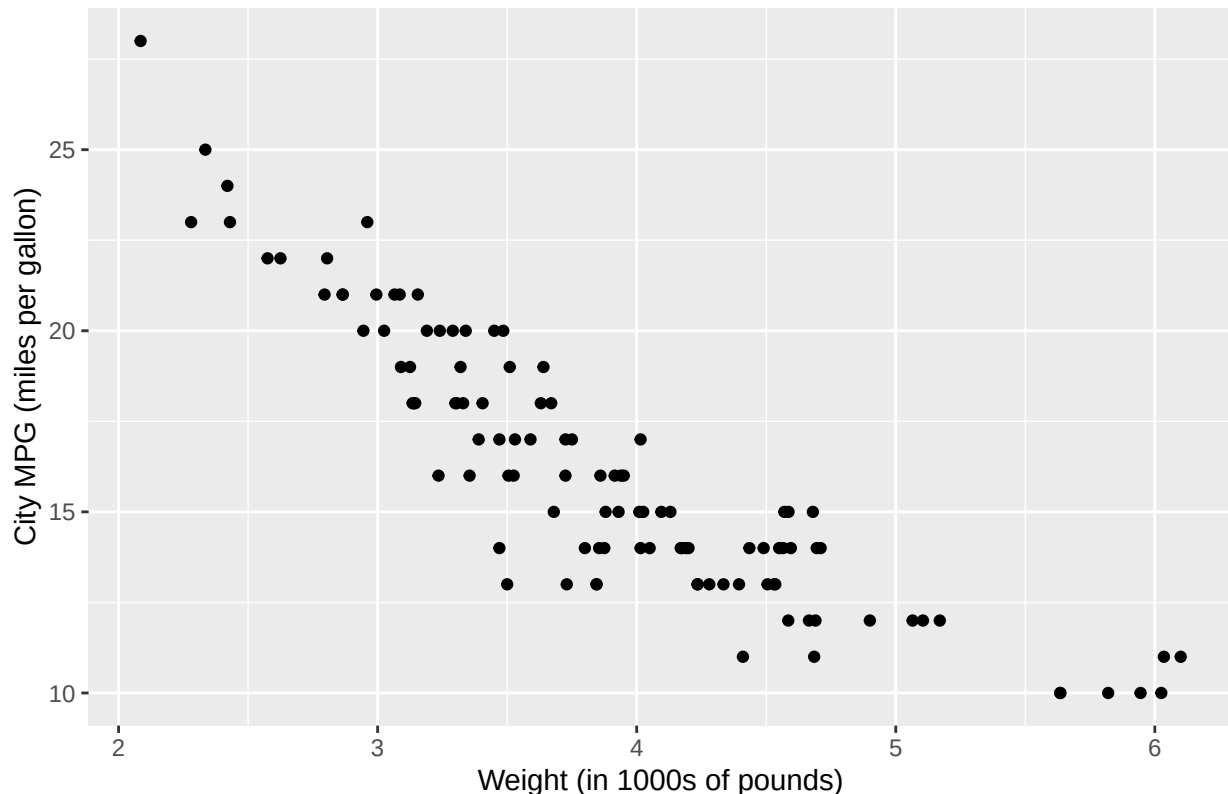
```
## # A tibble: 100 x 5
##   Make Model   Weight CityMPG weight_1000
##   <chr> <chr>   <dbl>   <dbl>     <dbl>
## 1 Acura MDX     4200     14       4.2
## 2 Acura RLX     3930     15       3.93
## 3 Audi  A3      3135     18       3.14
## 4 Audi  A4      3630     18       3.63
## 5 Audi  A6      4015     17       4.01
## 6 Audi  Q3      3880     15       3.88
## 7 Audi  TT      3140     18       3.14
## 8 BMW    3 Series  3640     19       3.64
## 9 BMW    5 series  3950     16       3.95
## 10 BMW   7 Series  4710     14       4.71
## # i 90 more rows
```

Quiz questions

1. Make a scatterplot of CityMPG versus weight_1000. Add appropriate axis labels to your graph. Be sure to include units of measurement.

```
ggplot(cars_2020, aes(x = weight_1000, y = CityMPG)) +
  geom_point() +
  labs(title = "Scatterplot of City MPG vs. Weight",
       x = "Weight (in 1000s of pounds)",
       y = "City MPG (miles per gallon)")
```

Scatterplot of City MPG vs. Weight



- Describe the relationship between the two variables that you see in the scatterplot as we've practiced in class. Be sure to address the strength, shape, and direction of the association you see, and use numbers taken from the graph to support your description. Look back at our previous work for the recipe here. Resist the temptation to "Nail It."

This scatterplot shows the relationship between city MPG and the weight of vehicles. The general shape of the relationship appears to be linear, with a strong association. This means that as the weight of a vehicle increases, its fuel efficiency in city driving decreases.

For example, a vehicle that weighs around 3,000 pounds achieves approximately 20 miles per gallon. In contrast, a heavier vehicle, weighing close to 5,000 pounds, only gets about 12 MPG. This trend suggests a negative direction meaning that heavier vehicles consume more fuel per mile when driving in city conditions.

- Create a linear model for the relationship between these two variables. You should do this even if the relationship does not appear to be linear! Report the formula for your model here.

```
# summary stats
model <- lm(CityMPG ~ weight_1000, data = cars_2020)

summary(model)
```

```
##
## Call:
## lm(formula = CityMPG ~ weight_1000, data = cars_2020)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.6328 -1.0471  0.0008  1.1300  4.7975
##
```

```
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  31.4094     0.7512   41.81  <2e-16 ***
## weight_1000  -3.9362     0.1894  -20.79  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.674 on 107 degrees of freedom
## Multiple R-squared:  0.8015, Adjusted R-squared:  0.7996
## F-statistic:  432 on 1 and 107 DF,  p-value: < 2.2e-16

CityMPG_hat = (-3.9362) * weight_1000 + (31.4094)
```

4. Write a sentence interpreting the slope of your model in the style of the sentences that we have used in class activities. Use the context of the data set in your sentence and include appropriate units of measurement.

The slope of -3.9362 means that for every increase of 1,000 pounds in vehicle weight, the predicted cityMPG decreases by approximately 3.94(rounded) miles per gallon. Which confirms the description from earlier that heavier cars tend to have lower fuel efficiency.

5. What does your model predict for the city mileage of a 2020 Toyota Camry? What is the residual?

$\text{CityMPG_hat} = (-3.9362) * 3.340 + (31.4094)$ The predicted cityMPG for the Camry is 18.26 MPG.

$\text{residual} = \text{actualMPG} - \text{PredictedMPG} = 20 - 18.26 = 1.74$ residual = 1.74

This means the model underestimated the camrys actual fuel efficiency by 1.74mpg.

6. Interpret the R^2 value from the model output in context, as we've done in class.

R^2 value is 0.8015 which means that 80.15% of the variability in city mpg can be explained by the weight of the vehicle(in 1,000s of pounds). This suggests that vehicle weight is a strong predictor of fuel efficiency in city driving. However since this is not 100% it is likely other factors also contribute to the city mpg such as if the car is gasoline, electric or a hybrid version.